

to adjust the watering schedule. What RainMachine sees, you can see on your smartphone or tablet. It is a Bluetooth-controlled low energy sensor that precisely measures in real time. It measures the parameters for plant growth, including soil moisture, fertilizers, ambient temperature and light intensity. It alerts you on your smartphone if any action is required, via a free and dedicated app.

Parrot's wireless sensor technology relies on a seamless connection with GreenIQ weather-based irrigation technology. It has more than 7000 plants in its database, which you can browse by type, shape, bloom, leaf colour, lifetime (annual, biennial, perennial) and season.

Edn Wall Garden is a soil-less vertical garden that waters itself (with nutrient-infused water) and monitors its own temperature with smart lighting. It lets you have up to 14 types of fresh herbs, vegetables and flowers growing easily in your abode. It automates tasks such as irrigation, lighting and dosing of nutrients.

Nano Garden's built-in lighting promotes plant growth, while drawing water from the house water supply.

Capacloud smart vertical garden offers hassle-free sustainability, zero maintenance, low operation and maintenance costs.

Practical benefits

Every time a state in India faces severe drought, it causes adverse economic impact on the agricultural sector. If the above-mentioned techniques are implemented, farmers do not have to remain dependent on climatic changes, lack of water and other challenges faced due to fluctuating moisture conditions.

Such systems work best for areas unsuitable for conventional farming, such as arid, degraded or deserted regions. These are independent of weather and, hence, allow cultivation

throughout the year.

Vertical gardens add to fast economic return, including optimal use of water and other resources. Thus, with increase in production, these give better standardisation for cultivation. These gardens make better use of land, thereby increasing total yield.

The best part about this technology is that the nutrients can be better controlled and monitored than the crops cultivated in soil. A disinfection system can also be installed to avoid infections from reused water.

In order to survive, we require air, food, water and living space. With growing population, we need to optimise land area, while preserving biodiversity. To meet up to the food demands of this growing population, it is necessary to implement vertical gardening. It gives optimum produce in a limited space.

Absorption in vertical gardening is usually proportional to concentration of nutrients in the solution near the roots. Productivity is influenced by environmental factors such as salinity, oxygenation, temperature, pH and conductivity of the nutrient solution, light intensity, photoperiod and air humidity. Due to the above characteristics—which enable an appreciable restriction in pesticide use and nutrient leaching—hydroponics is considered not only compatible but also strongly favourable and supportive to the application of integrated crop management. **EFY**



Parrot Flower Power works with GreenIQ technology
(Credit: pinterest.com)

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